

Ravi Ranjan Bhardwaj

Automation & Robotics Engineer

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Profile

B.Tech graduate in Automation & Robotics Engineering (GGSIPU, Delhi) with a Diploma in Electronics Engineering (IERT, Prayagraj) before that — background in circuits and hardware, extended into full-stack development and applied AI/LLM work. Experience spans solar tracking hardware, industrial robotic welding, IoT energy monitoring, and production dashboards, alongside leadership of university-scale clubs, fests, and hackathons.

Experience

Winsol Hybrid Power Pvt. Ltd. (TWI GROUP), IoT Engineer Aug 2026 - Present
New Delhi, India

- Building the solar-tracking module of an IoT monitoring platform for a **Solar + Wind Hybrid Power Plant**: real-time sensor acquisition, **ESP32 integration**, and cloud-connected dashboards
- Writing Python scripts for sensor data processing and automation, plus a **React.js dashboard** visualizing energy output and system health in real time

Airmount Logistics Pvt. Ltd., Technology & Supply Chain Operations Intern Feb 2026 - Aug 2026
New Delhi (Hybrid)

- Shipped **React.js UI components** for the internal logistics dashboard: responsive layouts, state management, and **Node.js/Express API** integration, **reducing page-load errors by ~30%**
- Migrated shipment records from spreadsheets to **MySQL** and wrote queries to automate shipment-tracking reports for the supply chain team

Innodata Inc., AI Data Annotator (Freelance) Jan 2026 - Feb 2026
Remote

- Annotated and validated **500+ data samples** for Generative AI/LLM fine-tuning: text classification, prompt-response evaluation, and quality ranking
- Flagged labelling inconsistencies and edge cases in **RAG pipeline** outputs, contributing to measurable improvements in model accuracy and response consistency

IBM SkillsBuild & CSRBOX, Agentic AI Architect Intern Jul 2025 - Aug 2025
Remote

- Built a **multi-agent AI system** for sensor monitoring, anomaly detection, and automated decision-making using Python and Generative AI
- Implemented **predictive maintenance logic** and adaptive control workflows aligned with **Industry 4.0** principles

Sunrise Engineering Company, Manufacturing Intern Jul 2025
Vadodara

- Hands-on exposure to manufacturing and assembly of forced lubrication systems, heat exchangers, oil filters, and filter elements
- Learned **quality inspection and testing procedures** on an active production floor

STEAM Varsity Makerspace (EdTech), Technical Intern Feb 2025 - Jan 2026
Gurugram (Hybrid)

- Designed and built **IoT prototypes and embedded systems demos** on Arduino/ESP32; wrote C++ and Python firmware for sensor interfacing and motor control
- Delivered hands-on workshops for **100+ students** covering drone assembly, automation logic, and full-stack IoT project development

Airmount Logistics Pvt. Ltd., Web Developer Intern Jul 2024
New Delhi (Hybrid)

- Built frontend features with **React.js** and JavaScript for the company's internal portal and fixed cross-browser UI bugs
- Handled **MySQL** data entry and table optimization for logistics tracking modules

Projects

AI Industrial Maintenance Assistant Python, Scikit-learn, LangChain, FAISS, Streamlit

- Predictive-maintenance dashboard for a **15-machine plant floor**; a **Random Forest model** scores failure risk from live sensor readings, and a **RAG chatbot** (LangChain, FAISS, Gemini) answers maintenance questions from that same sensor history

3D Scene Reconstruction & Mapping Python, Computer Vision, 3D Geometry

- Turned a single mobile photo into a measured 3D scene (table, walls, floor, **13 objects**) with real positions and sizes; camera calibration and scale cues validated against hand-taken ground truth at **90-98% accuracy**

Automatic TIG/MIG Robotic Welding System (Major Project) Python, Flask, React.js, OpenCV, Industrial Robotics

- Industrial robotic arm welding system with a **Flask/React control panel** for process parameters and an **OpenCV pipeline** for real-time weld-defect detection, cutting manual intervention and rework versus hand welding

CognoDB Graph Database Application (DepGraph) TypeScript, Neo4j, Cypher

- Supply-chain risk explorer that traces which services are reachable from a vulnerable package and through which dependency chain, over a **Neo4j graph** queried with **openCypher**

Go Auth CLI with 2FA Go, Docker, SQLite, TOTP

- Containerized CLI login system with **bcrypt-hashed passwords**, **TOTP two-factor auth** with a scannable QR code, account logout, and session timeouts, backed by SQLite in a Docker volume

PUMA 560 Robot Motion Simulation Python, PyBullet, Robotics

- GUI **PyBullet simulation** of a PUMA 560 arm that plans a quintic trajectory to a target X, Y, Z, checking joint limits, reachability, and self-collision; built for an **HTIC, IIT Madras** robotics assignment
- VectorShift Pipeline Builder

React, React Flow, Zustand, FastAPI

 - Visual, node-based pipeline builder for AI data workflows (input/LLM/transform/filter/merge nodes) validated as a **DAG** by a **FastAPI** backend before execution
- Smart Energy Monitoring System

ESP32, IoT, Blynk, Sensors

 - IoT electricity meter reading voltage, current, power, and energy via **ZMPT101B and SCT-013 sensors**, streamed live to a **Blynk dashboard** with EEPROM backup against power outages
- Programmable Robotic Arm on RC Car

Arduino, Servo Motors, Embedded C++

 - Joystick-controlled robotic arm with **6 servo motors** and **360-degree rotation**, mounted on an RC car chassis and driven over a two-channel transmitter-receiver link to an Arduino UNO
- Double Axis Solar Tracker

555 Timer IC, LDR Sensors, Motor Control

 - Dual-axis solar tracking rig on 555-timer and LDR logic with **L293D motor control**, built from scratch and tested through a full day to confirm improved energy capture over a fixed panel

EDUCATION

B.Tech in Automation and Robotics Engineering (Lateral Entry)	2023 - 2026
<i>University School of Automation & Robotics (USAR), GGSIPU, East Delhi Campus — First Division</i>	
Diploma in Electronics Engineering	2020 - 2023
<i>Institute of Engineering and Rural Technology (IERT), Prayagraj — First Division</i>	
Class 10 (CBSE)	2019
<i>Indian Public School, Madhubani, Bihar — First Division</i>	

TECHNICAL SKILLS

Programming & Web	Python, JavaScript, C++, Go, SQL, HTML/CSS, React.js, Node.js, Flask
Robotics, IoT & Automation	Arduino, ESP32, Embedded C/C++, Sensors & Actuators, PLC Basics, ROS, Industrial Automation
AI & Data	Generative AI, Agentic AI, LangChain, RAG, Computer Vision (OpenCV), Prompt Engineering, Data Annotation
Databases & Tools	MySQL, Neo4j, Git & GitHub, Docker, VS Code, MATLAB, Postman
Core Engineering	Analog & Digital Electronics, Control Systems, Manufacturing, Circuit Design, Supply Chain Ops
Design & Media	Adobe Photoshop, Premiere Pro, Canva, Photography

LEADERSHIP

University Student Representative	GGSIPU
Elected liaison between students, faculty, and administration; coordinated campus events and student activities	
President / Vice President	Anvaya Club, GGSIPU EDC
Led an interest-driven skill-building club: workshops and events across technical, entrepreneurial, creative, and photography tracks	
Office Head	Anugoonj 2025, Annual Cultural Fest, GGSIPU
Ran office operations for the university's flagship 3-day cultural fest: documentation, scheduling, and cross-team coordination	
Outreach Lead	GenAI Summit 2025
Led outreach, coordinated teams, and drove partnerships for the summit	
Media Head	Smart Delhi Hackathon 2025 & ICAPIE 2024
Ran media, content, and event coverage for a state-level hackathon and an international engineering conference	
Core Team	Elysian Tech Fest & SPIC MACAY
Supported logistics, participant engagement, and on-ground execution for the annual tech fest and cultural heritage events	

ACHIEVEMENTS & CERTIFICATIONS

- IBM SkillsBuild: Agentic AI Architect Certification
- Student Ambassador, IBM SkillsBuild Agentic AI Program
- Deloitte (Forage): Data Analytics, Cybersecurity & Technology Job Simulations
- TCS iON: Communication Skills Certification
- Letters of Recommendation: GenAI Summit 2025, Smart Delhi Ideathon 2025
- Certificate of Appreciation (No. 362): Office Head, Anugoonj 2025, GGSIPU
- 2nd Position, Science Quiz, National Science Day, GGSIPU

LANGUAGES

Hindi (Native), Maithili (Native), English (Professional), Bhojpuri (Elementary)